

TELECOM CUSTOMER RETENTION · MACHINE-LEARNING ANALYSIS

Why Customers Leave and How to Keep Them

Insights and evidence-based retention recommendations from a churn analysis of 7,043 telecom customers.

7,043

customers analysed

26.5%

overall churn rate

6

models compared

3

behavioural segments

EXECUTIVE SUMMARY

One in four customers churns, but the risk is highly concentrated

Churn is not spread evenly across the base. It clusters around a handful of contract, tenure, pricing and service conditions, which makes it targetable. This analysis identifies those conditions, ranks their predictive strength, and translates them into a prioritised retention plan.

THE PRIZE

5%

more retention can lift profit by
25–95%.

Acquiring a new customer costs 5–25× more than keeping one, so cutting churn is the highest-leverage growth lever available. ¹



Contract

Month-to-month customers churn at 42.7% vs 2.8% on two-year terms, the single strongest signal.



Tenure

53% of customers churn in their first six months; risk falls sharply with age on network.



Segments

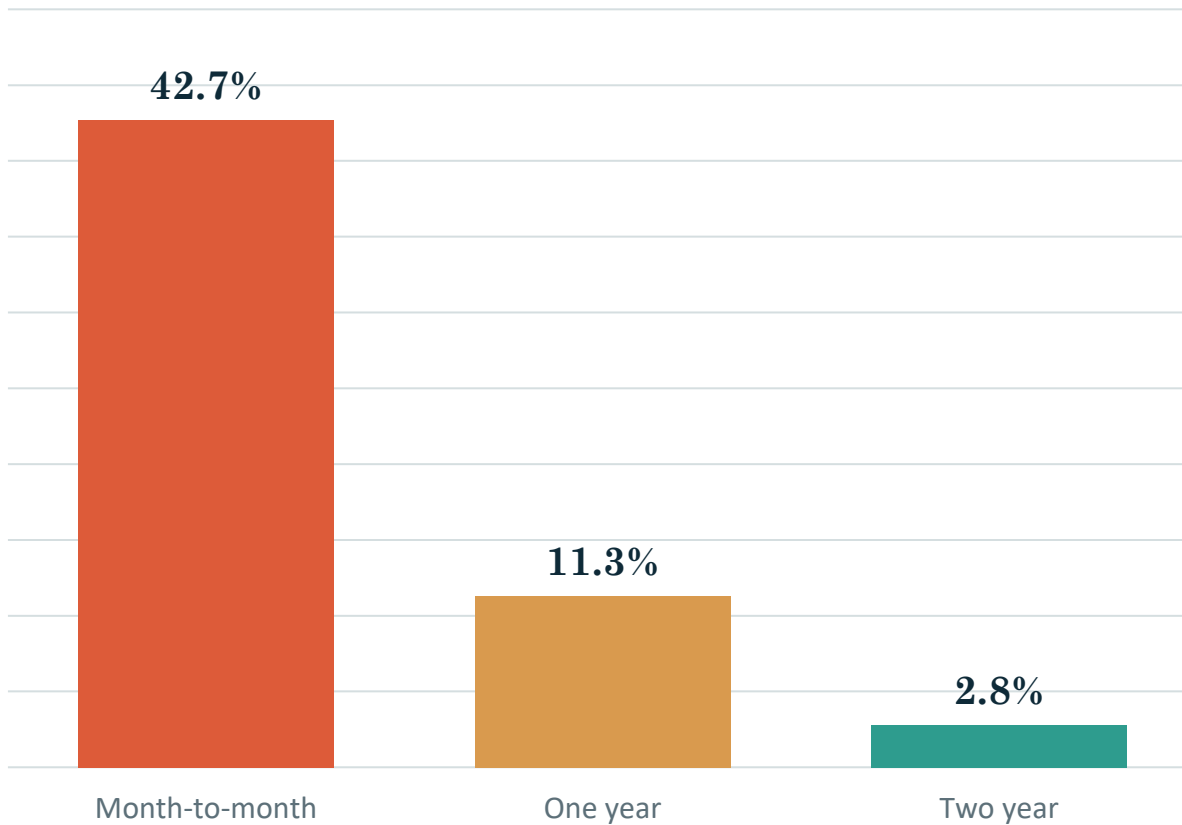
A 'new, high-spend' segment (34% of the base) churns at 47%, the clear priority group.



Prediction

Logistic regression reaches 0.85 ROC-AUC, giving a workable early-warning score.

Contract type is the dominant driver of churn



Churn rate by contract type

15×

higher churn on month-to-month plans than on two-year contracts.

- **Highest association of any variable**

Cramér's V = 0.41 against churn, the top-ranked categorical predictor.

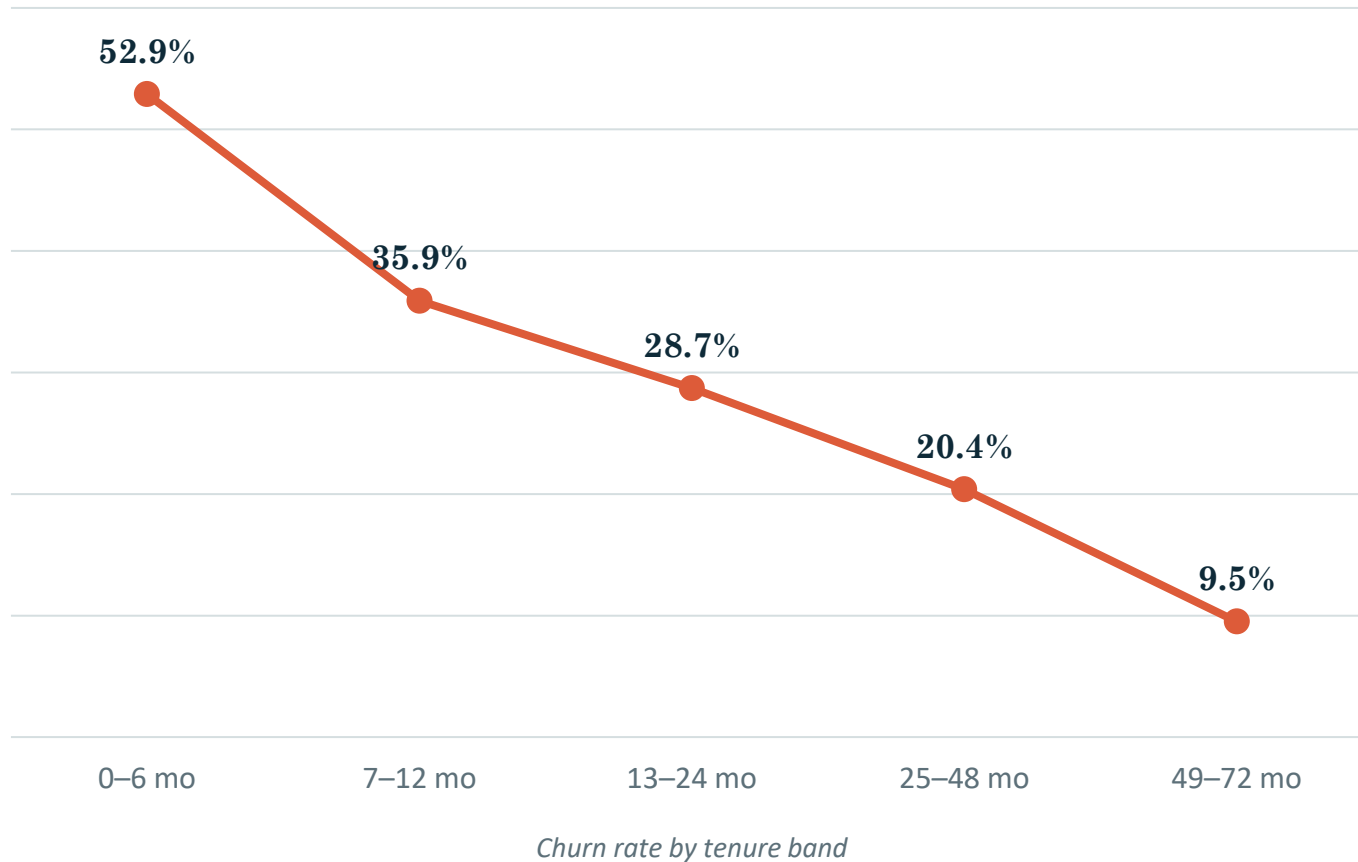
- **It is also the biggest exposure**

Month-to-month plans make up 55% of the customer base.

- **Commitment changes behaviour**

Each step up in contract length roughly quarters the churn rate.

The first year is where customers are lost



47.4%

of customers churn within their first 12 months

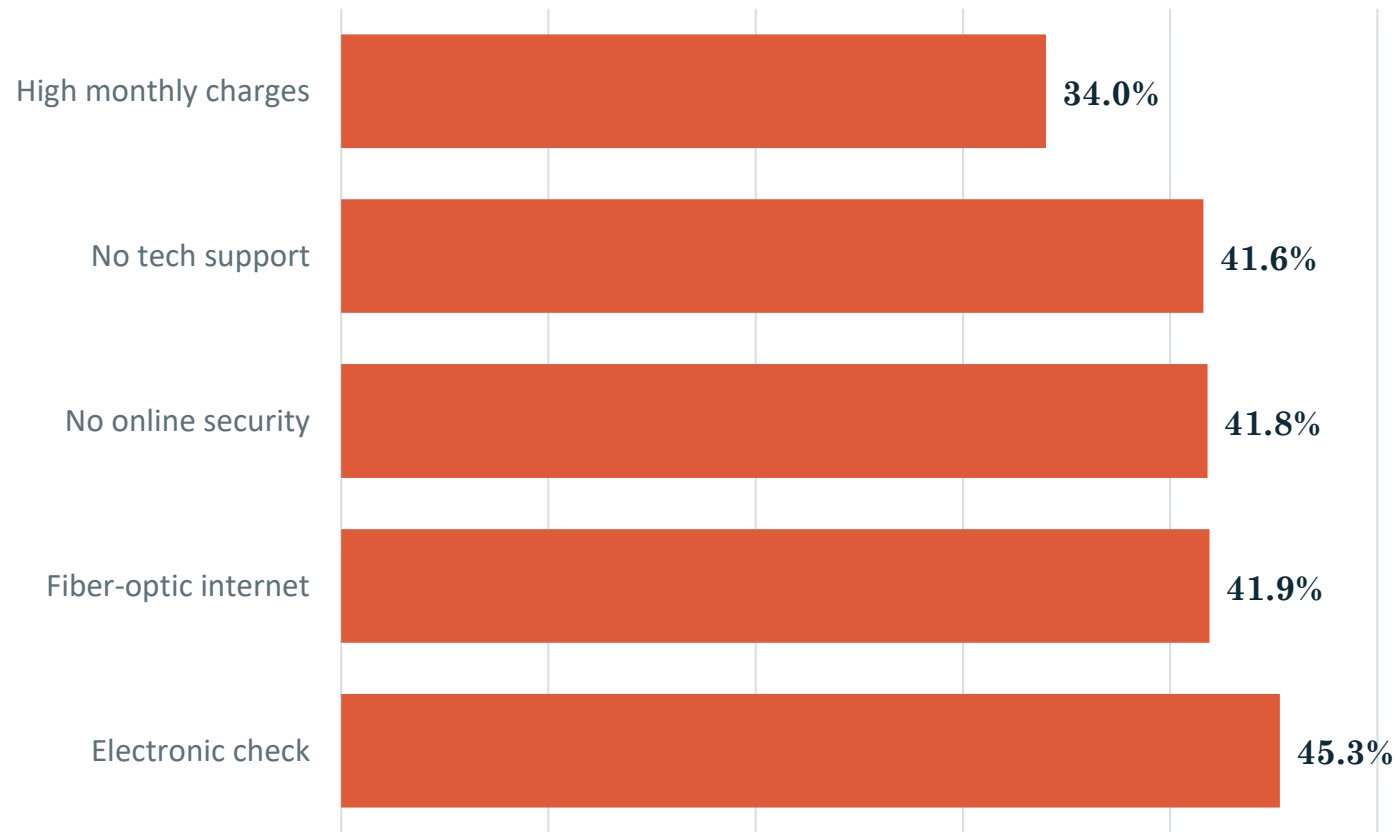
10 mo

median tenure of a churner, vs 38 months for those who stay

$r = -0.35$

tenure is the strongest numeric predictor of churn

Specific products and payment methods concentrate risk



Baseline churn across all customers: 26.5%

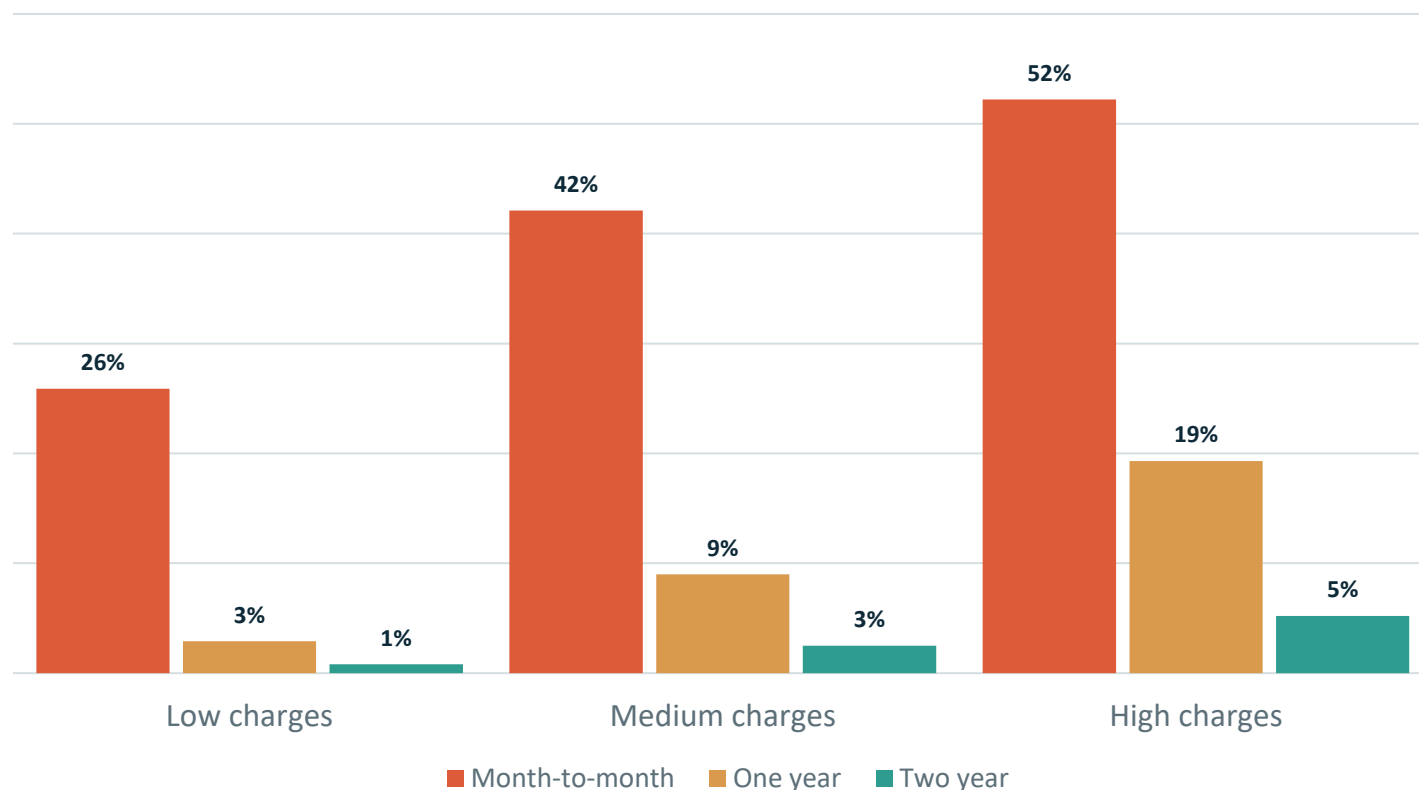
Payment friction

Electronic-check users churn at 45%, nearly 3× the 16% seen on automatic bank or card payments. Manual payment is a recurring exit ramp.

Missing 'sticky' add-ons

Customers without online security or tech support churn at ~42% vs ~15% with them. Fiber-optic, the premium tier, shows the highest churn of any internet type, signalling a price-to-value gap.

Price and contract risk stack on top of each other



Churn rate by monthly-charge band × contract type

HIGHEST-RISK CELL

52%

High-charge, month-to-month customers, over half leave.

LOWEST-RISK CELL

0.8%

Low-charge, two-year customers are effectively locked in.

Clustering reveals three segments, one is the priority

AT RISK

New high-spend

~15 mo · \$79/mo

47%

churn rate

34% of customer base**STABLE**

Budget low-spend

~27 mo · \$30/mo

16%

churn rate

36% of customer base**PROTECT**

Loyal high-value

~59 mo · \$91/mo

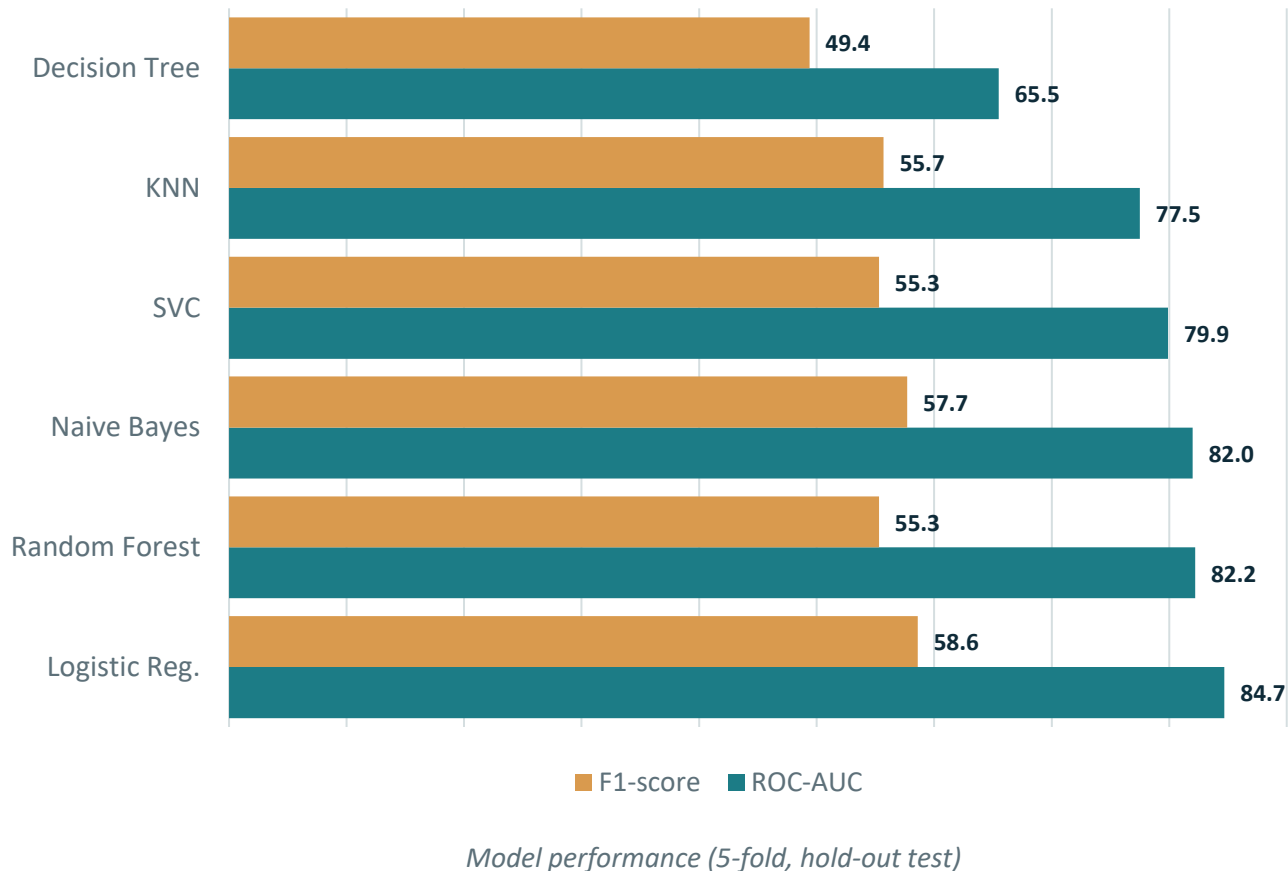
15%

churn rate

30% of customer base

The 'new high-spend' segment pairs the largest slice of the base with the highest churn, the single most valuable group to retain.

Churn is predictable, with a precision/recall trade-off



BEST OVERALL

Logistic Regression

0.85 ROC-AUC · 80% accuracy · interpretable and easy to deploy as a risk score.

THE CATCH: RECALL

At its default threshold the model catches only ~52% of churners. Retention needs the opposite bias.

Fix: lower the decision threshold (or use the recall-oriented model, which reaches ~85% recall) so fewer real churners slip through, accepting more false alarms is cheap next to a lost customer.

FROM INSIGHT TO ACTION

Five retention moves, prioritised

Each recommendation is anchored to a finding above and, where it draws on established retention research, cited.

- 1 Convert month-to-month to term contracts
- 2 Own the first 90 days with proactive onboarding
- 3 Launch a save programme for 'new high-spend'
- 4 Remove payment & value friction
- 5 Deploy the churn model as an early-warning score

Highest impact

Highest impact

High impact

Quick win

Enabler

Lock in commitment, win the first months, save the at-risk

1

Convert month-to-month to term contracts

Target high-charge month-to-month customers with incentives to move to annual plans, loyalty pricing, bundled perks, or a bill credit. The data shows each step up in contract length roughly quarters churn; commitment mechanics are among the most reliable retention levers.¹

EVIDENCE

MTM 42.7% → two-year 2.8% churn

2

Own the first 90 days with structured onboarding

Nearly half of churn happens in year one, so shift spend from acquisition to onboarding: proactive welcome contact, service-works confirmation, first-bill explanation, and early value milestones. Early-tenure churn is largely an onboarding problem, not a price problem.^{2 3}

EVIDENCE

First 12 months = 47% of all churn

3

Stand up a save programme for 'new high-spend'

Score this 34%-of-base segment weekly and intervene before intent hardens, the closer outreach is to the decision, the higher the save rate. Pair with right-sizing plans and add-on bundles to close the price-to-value gap.^{4 5}

EVIDENCE

47% churn in the priority segment

Superscripts (1–5) refer to the sources on the References slide.

Clear the friction, then let the model do the watching

4

Remove payment and value friction

Migrate electronic-check payers to autopay with a small incentive (autopay users churn at 16% vs 45%). Bundle online security and tech support into mainstream plans, and review the fiber-optic price-to-value gap. ⁴

EVIDENCE

Autopay 16% vs e-check 45%

5

Deploy the model as an early-warning score

Operationalise logistic regression as a weekly risk score, tuned for recall so few real churners are missed, feeding the save programme. Proactive scoring beats reactive win-back, intervene before the decision is made. ⁵

EVIDENCE

0.85 ROC-AUC early-warning

WHY IT PAYS OFF

5–25×

cost of acquiring vs retaining a customer ¹

25–95%

profit lift from a 5-point retention gain ¹

55%

of the base (month-to-month) is addressable now

47%

churn in the segment the plan targets first

REFERENCES

Sources for the cited recommendations

Data findings and segment figures are derived directly from the analysis of the 7,043-customer dataset and are not externally cited. The sources below support only the recommendations that draw on established retention research.

- 1 Reichheld, F. F. & Sasser, W. E. (1990). “Zero Defections: Quality Comes to Services.” Harvard Business Review. Establishes that a ~5% retention gain raises profit 25–95% and that acquisition costs 5–25× retention (Bain & Company).
- 2 Onboarding & first-90-day churn research (ChartMogul / subscription-churn benchmarks, 2025–26): the majority of early churn is determined in the first 30–90 days, before renewal.
- 3 Sequential Tech (2026), “Telecom Customer Onboarding Services to Reduce Early Churn”: proactive onboarding (service confirmation, first-bill explanation) reduces early churn, largely an onboarding, not price, problem.
- 4 CMSWire (2026), “The Real Customer Churn Problem: You’re Measuring It Too Late”: payment behaviour is an early churn predictor; flexible/automatic payment options stabilise accounts.
- 5 Databricks (2026), “Why Telecom Churn Prediction Misses the Intervention Window”; and Sustainability 18(3):1400 (2026), proactive churn retention via Lean Six Sigma: intervene early enough to change the outcome; save rate rises the closer outreach is to the decision.